

One-dimensional multipulse laser machining of structural alumina: evolution of surface topography

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Abstract The development and understanding of laser–material interactions have steered to the machining of advanced structural ceramics. At one point, it was nearly impossible to machine effectively using various conventional machining techniques. Nevertheless, achieving a higher material removal rate along with a good surface finish is a critical issue. In this study, a multistep computational model based on COMSOL™ Multiphysics was designed and developed to study the influence of multiple laser pulses on the evolution of surface roughness of alumina. The computational model employed the various heat transfer and hydrodynamic boundary conditions and thermomechanical properties for better prediction of surface roughness under various laser processing conditions. The results indicate that, as the pulse rate increases, the surface roughness also increases. The results of the computational model are also validated by experimental observations with reasonably close agreement.

Keywords Nd:YAG laser · Laser machining · Multiple pulses · Alumina · Computational model · Surface roughness

Abbreviations

Nd:YAG	Neodymium-doped yttrium aluminum garnet
CO ₂	Carbon dioxide
ρ	Density (in kilograms per cubic meter)
C_p	Specific heat at constant pressure (in joules per kilogram per kelvin)
k	Thermal conductivity (in watts per meter per kelvin)
T	Temperature (in kelvin)
t	Time (in seconds)
x	x -coordinate
y	y -coordinate
δ_m	Normalization function around the melting temperature (per kelvin)
δ_v	Normalization function around the vaporization temperature (per kelvin)
H'	Smooth Heaviside function
B	Fraction of a particular phase (solid, liquid, or vapor)
T_x	Transition temperature between two distinct phases, solid–liquid or liquid–vapor (in kelvin)
mod	Modular function
u	Velocity in the x -direction (in meters per second)
v	Velocity in the y -direction (in meters per second)
p	Pressure (in newtons per square meter)
T_s	Instantaneous surface temperature (in kelvin)
SiO ₂	Silicon dioxide
Fe ₂ O ₃	Iron oxide
R ₂ O	Base or fluxing oxides
μm	Micrometer or micron (10^{-6}m)
R_t	Maximum height of surface profile (in micrometers)
R_p	Maximum peak height of surface profile (in micrometers)

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R_v Maximum valley depth of surface profile (in micrometers)

1 Introduction

Alumina, zirconia, magnesia, silicon carbide, and silicon nitride are grouped as structural ceramics to be considered as one of the most versatile groups of materials because of its high temperature stability, ability to resist deformation at elevated temperature, excellent wear and thermal shock resistance, chemical inertness, superior electrical properties, and lower density [1–4]. Therefore, structural ceramics have found applications in a variety of industries, including automobile, aerospace, medical, printing, textile, and electronic [1–4]. However, due to their low fracture toughness ($\sim 3\text{--}5\text{ MPam}^{1/2}$) and high hardness ($\sim 1,200\text{--}2,200$ Knoop hardness), they are difficult to machine using conventional machining techniques, such as grinding, cutting, and polishing [5–10]. Moreover, numerous other investigators [11–16] reported that these machining techniques are also associated with many drawbacks, such as unacceptable tool wear, insufficient dimensional accuracy, mechanical or thermal damage to workpieces, lower material removal rates, and surface and subsurface cracks.

Many researchers [17–21] have demonstrated that recent development of lasers have driven the machining of advanced structural ceramics, which previously were nearly impossible to machine effectively using various conventional machining techniques. Particularly, a one-dimensional laser machining technique (both the workpiece and the laser beam are stationary) has been successfully implemented in many industrial applications [21–24], such as aerospace (cooling holes in nozzle guide veins and gas turbines), automotive (fuel injection nozzles), and electronic (circuit boards). One-dimensional laser machining is inherently associated with several defects and many dimension-related issues. Some of the prominent issues reported are the formation of the recast layer on the walls of machined surface, spatter formation on the top surface, nonparallel machined walls, lack in dimensional accuracy, larger heat-affected zones, and microcrack formation [21, 25, 26]. Considerable amount of experimental and computational works [1, 2, 14, 15, 27–30] have been published in the past to improve the quality of laser machining by employing various types of lasers: CO₂, excimer, and Nd:YAG. These lasers can be operated either by continuous wave (CW) or pulse mode (PM) (millisecond, nanosecond, picosecond, and femtosecond lasers). In CW, the output energy of the laser beam is constantly ON for a specific amount of time, whereas in PM, the lasers concentrate their output energy into shorter-time high-power bursts. The PM lasers can either fire a single pulse or a series of pulses (multiple) at regular intervals. In both cases, the instantaneous power densities can be extremely high ($\sim 10^7\text{ J/m}^2$), and as a result, a larger volume of material is removed [1, 2]. In

the past, Samant and colleagues [1, 2, 14, 27, 28] used the experimental and computational approaches to understand the mechanisms of material removal during one-dimensional laser machining to various ceramics (alumina, zirconia, magnesia, silicon nitride, and silicon carbide). Their studies show that the higher material removal rate can be achieved by using multiple laser pulses and considered as a preferable machining technique for structural ceramics [1, 2, 8, 26, 31]. However, the main focus of their works was to increase the material removal rates by selecting the optimal laser processing parameters and disregarded the issues related to surface finish. Hence, the undesirable surface finish produced during one-dimensional laser machining is still one of the critical issues to be resolved.

In light of this, the present authors have already developed the computational model based on COMSOL™ Multiphysics validated with experimental trials to understand the influence of a single laser pulse on laser machining of alumina and its subsequent effect on surface finish [32]. It was observed that the surface roughness increases with increase in laser energy density during the application of a single laser pulse. In extension of this work and as part of the ongoing incremental efforts to develop a laser-based multidimensional (3D) machining through understanding of the basic science behind the machining phenomena, the present paper investigates the effect of multiple laser pulses on the evolution of surface topography during one-dimensional laser machining (stationary workpiece and laser beam) of alumina using both computational and experimental approaches. The computational model is developed to simulate the multiple pulses and then validated with the experimental observations to study the effects of various process parameters on the evolution of surface topography of the laser-machined alumina.

2 Laser–material interactions

The temperature rise and fall due to heating (absorption of the laser beam) and cooling (self-quenching by bulk material and heat losses due to external natural convective cooling and radiation) control different physical phenomena in the material during thermal processing. When a high-intensity laser beam ($\sim 10^6\text{ J/m}^2$) strikes the surface of a material, the surface temperature quickly rises due to heat conduction. This rapid rise in surface temperature leads to surface vaporization and the subsequent release of neutral atoms or molecules in a gas cover adjacent to the laser–material interaction zone [29, 33–35]. Furthermore, the evolving vapor from the melt pool is significantly cooler and denser than the atmosphere, which results in a condensation of vapor particle back on the surface which generates a recoil pressure on the melt pool underneath [29, 33–35].

The commonly used pulse width for laser machining is 0.1 to 10 ms [1, 2]. These time scales allow the surface to heat to

the vaporization temperature and remain there for some time [1, 2]. For higher laser energy densities ($\sim 10^6 \text{ J/m}^2$) and shorter pulse durations (0.1 to 10 ms), the resulting recoil pressure may be extremely high, which induces a shock wave. This generates a hydrodynamic melt motion on the liquid melt pool, as proposed by various investigators [29, 33, 34]. As a consequence, the liquid metal is ejected out from the laser–material interaction zone. The temperature of the expelled molten material drops drastically as soon as it leaves the laser–material interaction zone. Furthermore, the effect of recoil pressure on the molten material also nullifies as the surface temperature goes below vaporization temperature. At the end of a laser pulse, the surface tension and gravitational forces drive the liquid material to return to its original place. Due to self-quenching effect and a higher cooling rate ($\sim 10^5 \text{ K/s}$), the liquid material solidifies instantaneously and forms a liquid crown or liquid pileup. Further, Dahotre and colleagues [1, 2, 14, 15] reported that the tangential stress exerted due to surface tension gives shape to the solidified material. Thus, a pronounced surface topography is generated on the laser-machined surface (Fig. 1).

It was previously observed that, during the application of a single pulse ($\sim 10^6 \text{ J/m}^2$), the laser–material interaction zone ($\sim 10^{-10} \text{ m}^3$) reaches an extremely high temperature ($> 8,000 \text{ K}$) within a very short period of time ($< 0.5 \text{ ms}$) [32]. Therefore, it makes the situation enormously difficult for in situ measurements of thermophysical properties under these severe conditions (high temperature, short time of interaction, and small interaction volume). In consequence, it is extremely challenging to investigate the evolution of surface topography through experimental analyses. Hence, the computational modeling approach (via finite element method) is successfully used in the present study.

3 Modeling approach

For PM lasers, one-dimensional laser machining is carried out by either supplying a single laser pulse or multiple laser pulses by keeping both workpiece and

laser beam stationary. The average laser energy density (energy per pulse/surface area of laser beam) in each case is very high ($\sim 10^6 \text{ J/m}^2$), which results in a high loss of material. Similarly, by delivering an increased number of pulses, higher material removal rates can be achieved. In order to obtain higher material removal rates, PM lasers deliver multiple laser pulses (which is preferred for laser machining of structural ceramics). Therefore, in the present study, the effect of multiple laser pulses on the evolution of surface topography is modeled to understand the laser machining mechanism for various conditions. Subsequently, this understanding can be extended to predict the surface topography during multidimensional laser machining processes.

In PM laser machining, the multiple laser pulses of short pulse width are delivered in a repeated manner (Fig. 2a, b). The material removal during laser machining is carried out either by evaporation or by melt ejection, which solely depends on the laser machining parameters, such as average energy density, pulse width, and pulse rate (repetition rate). The laser (or heating) is ON (Fig. 2b) during the application of first pulse (till time t_1) with preset value of pulse width followed by cooling until the end of the first pulse (time $t_2 = 1/f$, in second). The surface temperature increases from initial temperature (T_i) to T_1 during heating, while during cooling, the temperature decreases from T_1 to T_2 (where $T_1 > T_2 > T_i$). Subsequently, the second pulse is supplied from time t_2 to t_3 and again cooling until time t_4 . This pulse train successively delivers a specific number of pulses over the entire time of machining operation (until time t_f). However, during the application of multiple laser pulses, the temperature at the beginning of every pulse was always higher than the initial temperature (T_i), causing a gradual rise in average surface temperature (Fig. 2b). Furthermore, the variations in surface temperature, in turn, influence the mechanism of evolution of surface roughness (topography). Hence, the distinct surface topography is generated (Fig. 2c) due to the impact of multiple laser pulses based on the laser pulsing parameter (Fig. 2d).

A two-dimensional axisymmetric computational model, which incorporates fluid flow and heat transfer, is developed using COMSOL™ Multiphysics (Fig. 2c). The model is used to predict the temperature, cooling rate, and fluid velocity (all of which influence the surface topology of alumina) after the surface is subjected to various laser machining conditions. A two-step modeling approach (Fig. 3) is used in the present computational model to predict the surface deformation under the application of multiple laser pulses. In step 1, the multi-phase (solid–liquid–vapor) heat transfer model is incorporated to evaluate the material loss due to evaporation which is used to predict the geometry of a crater.

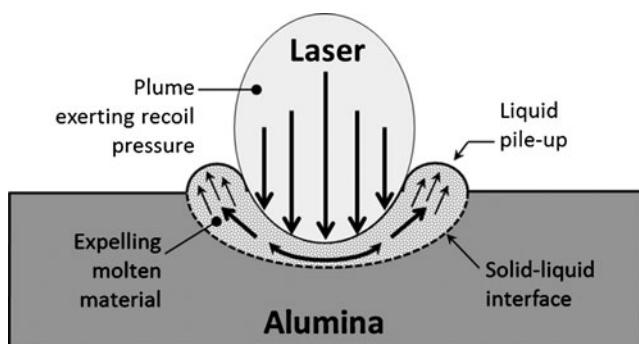
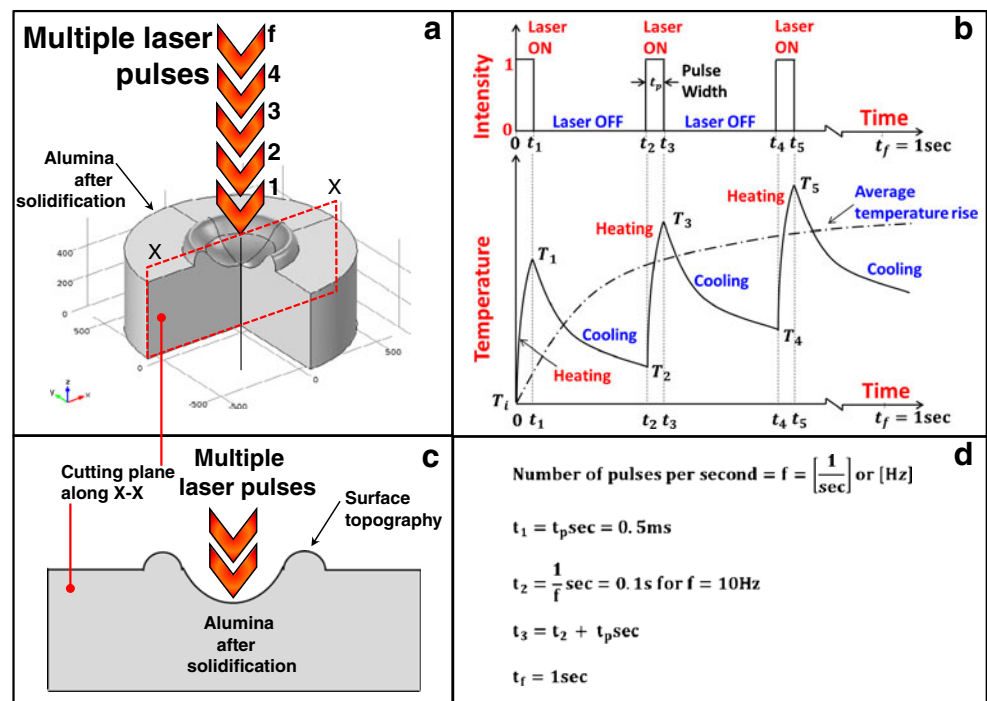


Fig. 1 Schematic of laser–material interaction and associated surface effects

Fig. 2 Schematic of laser machining processes: **a** multiple-pulse laser machining process, **b** temperature as a function of time, **c** evolution of surface topography, and **d** laser pulsing parameters



The initial geometry used in the step 1 computational model is presented in Fig. 4. In order to get the geometry-independent solution, a mesh sensitivity analysis was carried out for six different mesh sizes and the optimal mesh size was used for the computation (Fig. 3). The level set method is also employed in step 1 to

identify liquid–solid and vapor–liquid interfaces. In step 1, the elements identified as vapor phase are excluded from the geometry. The geometry of the crater (depth and width) evaluated from step 1 plays an important role in predicting the final surface profile. Therefore, the crater geometry predicted from step 1 is considered as

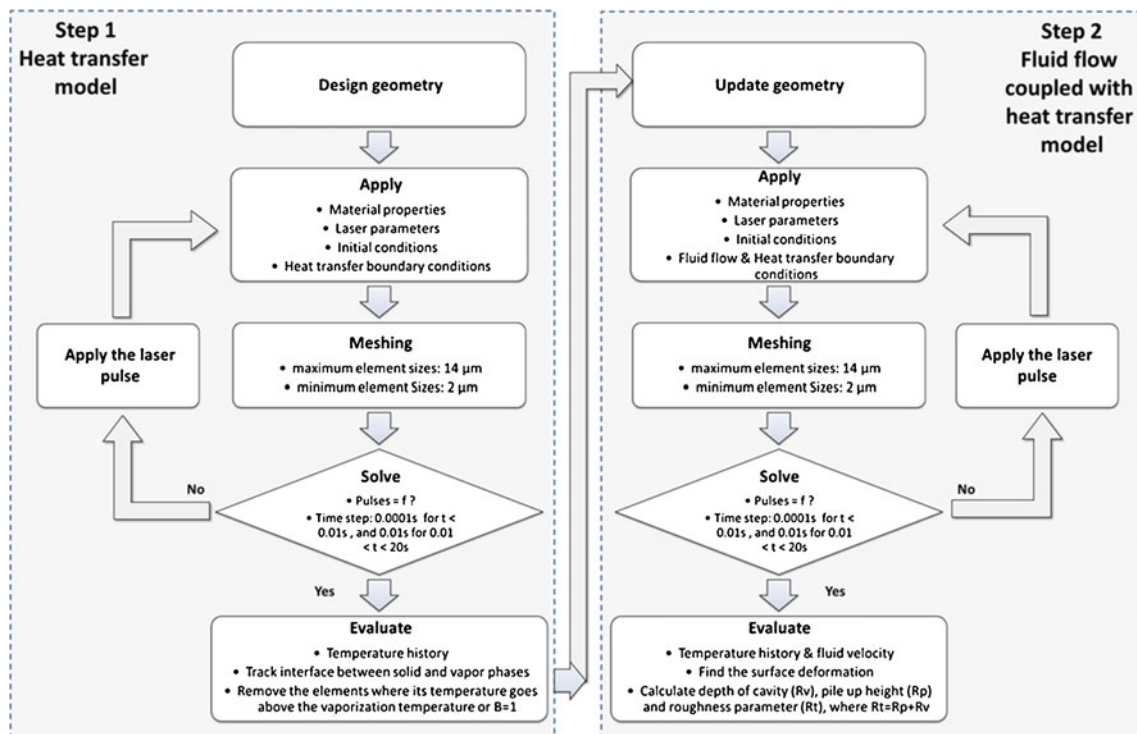


Fig. 3 Flow diagram for multistep computational modeling approach

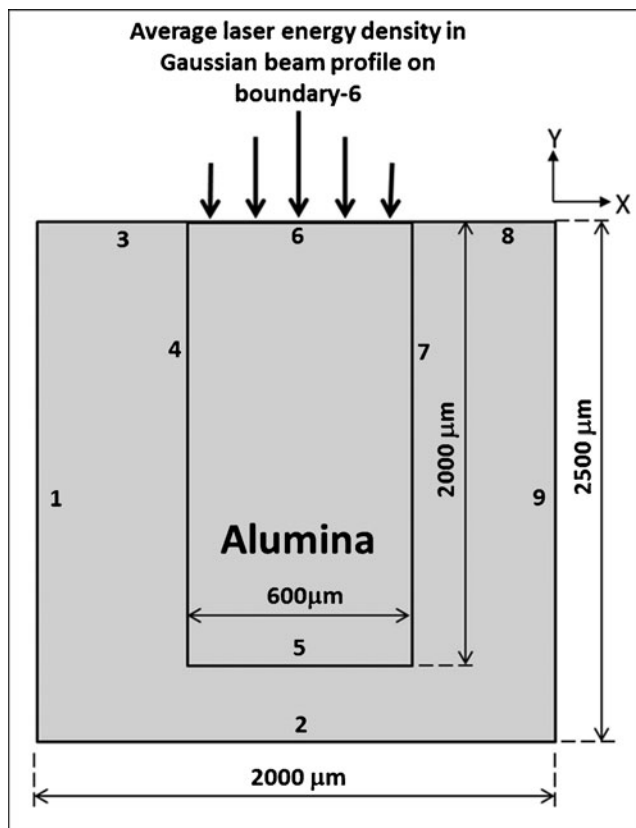


Fig. 4 Two-dimensional geometry for the computational model

an initial surface profile for step 2. In step 2, the multiphase fluid flow model coupled with heat transfer and phase change kinetics is employed. The prime objective of step 2 is to estimate the contribution of the deformation of the liquid material pool inside of the crater under various forces (recoil pressure, surface tensions, and gravitational forces) and to predict the final surface topography. This integrated computational approach involves various physical phenomena for predicting the surface profile during multiple-pulse laser machining of alumina (Fig. 5).

3.1 Heat transfer model

During laser machining, the material undergoes phase change (solid to liquid to vapor or vice versa). In addition, the heat (latent heat of fusion and evaporation) consumed or released during the phase transition significantly influences the magnitude of surface temperature, recoil pressure, melt pool size, and fluid velocity. Therefore, the temperature-dependent properties of alumina, such as density, thermal conductivity, and specific heat, are incorporated in the computational model. The change in density and thermal conductivity with respect to temperature (ranging from 293.15 to 3,500 K) is illustrated in Fig. 6. In addition, Table 1 summarizes

the material properties of alumina [1, 2, 36–39] and the laser machining parameters used in the computational model. The specific heat related to phase transition is used (C_p term) in the computational model, which is expressed by Eqs. 1–3 in Table 2 [40]. The governing equation and the various boundary conditions employed in the heat transfer model are also summarized in Table 2.

3.1.1 Simulation of multiple laser pulses

In the present work, multiple pulses (10, 20, 30, 40, and 50) are delivered in durations of 1 s. Thus, the analytic function (β) was designed to simulate the repetition of multiple laser pulses that are expressed by Eq. 7 enumerated in Table 2 [41]. As discussed earlier (Fig. 2b), during the application of the first pulse, the laser is ON with a specific value of pulse width ($P_w = 0.5$ ms) followed by cooling until the end of first pulse (time $t_2 = 1/f$, in seconds). Hence, the laser ON time is simulated by the rectangle function (lower and upper limits of 0 and 0.5 ms, respectively). The β function only works when $t \geq 0$ s and $t \leq 1$ s, so that it simulates the pulse train of 1 s.

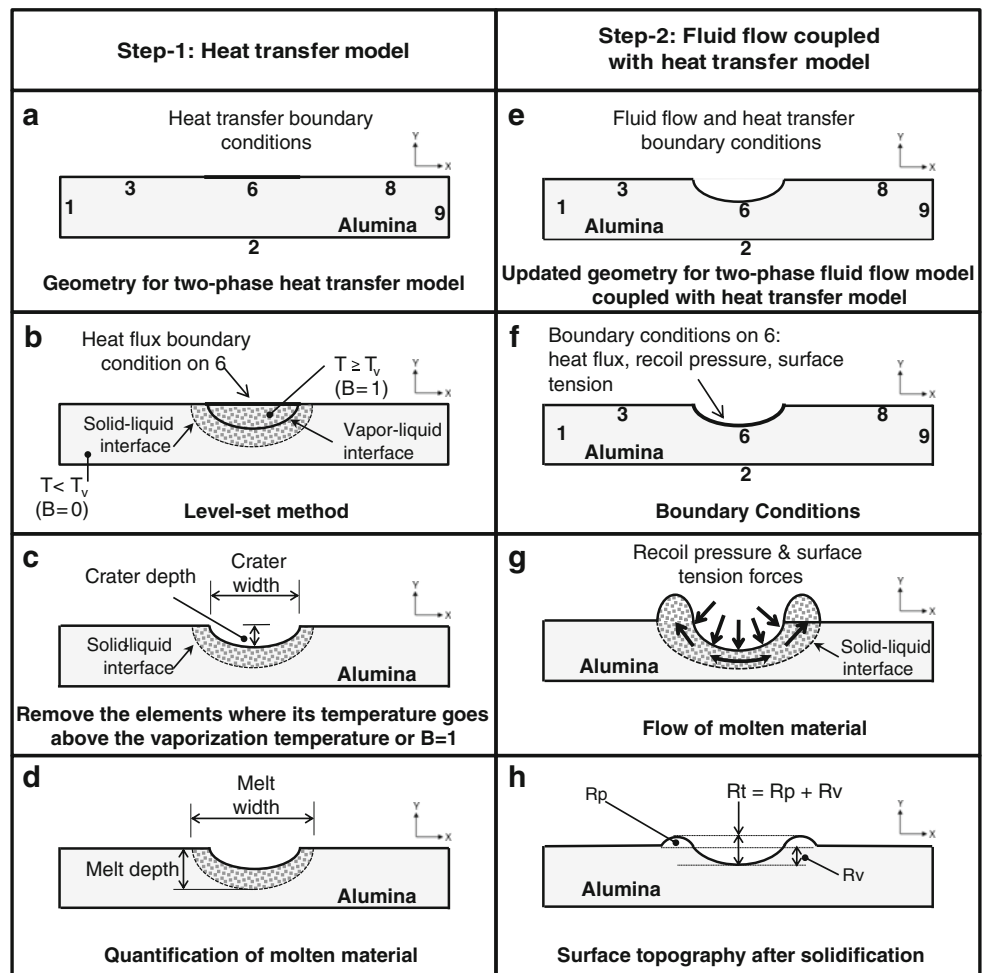
3.1.2 Prediction of crater geometry

The level set method developed by Sussman et al. [42] is employed in step 1 in order to track the solid–liquid and vapor–liquid interfaces as expressed by Eqs. 11–13 in Table 2. For alumina, T_x can be T_v to distinguish the vapor phase from the liquid phase and T_m for the liquid phase from the solid phase. The B ranges from 0 to 1. For $T_x = T_m$, $B = 1$ indicates 100 % liquid phase and $B = 0$ designates 0 % liquid phase. The interphases is tracked between $B = 0$ and $B = 1$. Similarly, for $T_x = T_v$, $B = 1$ indicates 100 % vapor phase and $B = 0$ designates 0 % vapor phase. The elements coming under vapor phase ($B = 1$ or $T > T_v$) are identified and removed from the geometry and the dimensions of the crater geometry are estimated which is further considered as an initial geometry for step 2 (Fig. 5).

3.2 Fluid flow coupled with heat transfer model

Multiphase fluid flow coupled with heat transfer and phase change kinetics is modeled in step 2. The new geometry is drawn by considering the material loss due to vaporization from the step 1 results. The governing equations for conservation of mass, momentum, and energy for two-dimensional transient, incompressible laminar flow coupled with heat transfer are expressed in Eqs. 14–17 listed in Table 3. The natural convection induced by the buoyancy-driven flow and volume force

Fig. 5 Schematic of a physical model in two-step computational modeling approach to predict the surface topography



is estimated by the Boussinesq approximation presented by Eq. 16 in Table 3 [43, 44]. Step 2 also utilized the same boundary conditions of heat transfer and phase change kinetics as used in step 1 (expressed by Eqs. 2–10 listed in Table 2).

The evaporation-induced recoil pressure (P_r) at the evaporating surface depends on the incident laser energy density as expressed by Eq. 18 in Table 3 [1, 2]. In the present model, the experimentally verified physical model of melt hydrodynamics and laser-induced evaporation proposed by Anisimov [33] and Semak [34] are used to simulate the effect of recoil pressure. The temperature fields determined from the computational model facilitate the calculation of evaporation-induced recoil pressure at the evaporating surface during laser machining of alumina. In the present model, the recoil pressure is applied only when the temperature is above the vaporization temperature ($T \geq T_v$).

The forces at the liquid–air interface due to Marangoni convection are incorporated in such a way that the shear stress is proportional to the temperature gradient on the surface [43]. In order to apply this condition, the weak term is added from “weak contribution” feature of the laminar

flow interface shown by Eq. 19 in Table 3. This term imposed the velocity in the x -direction and overrides the slip boundary condition. In the laser–material interaction zone, where the temperature is at its highest, the surface

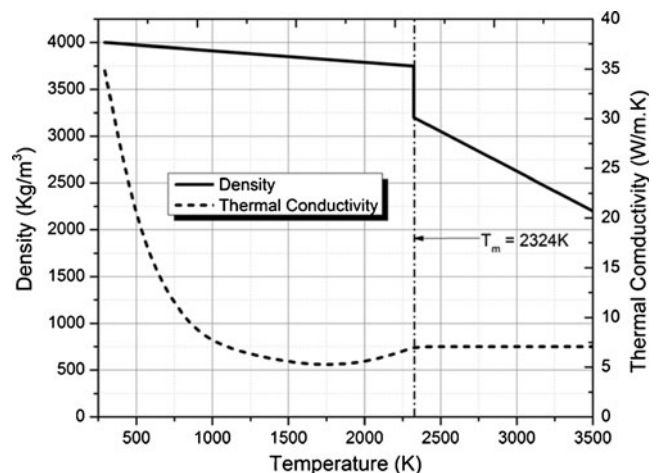


Fig. 6 Temperature dependent material properties of alumina density and thermal conductivity

Table 1 Material properties of alumina (99.6 %) and laser machining parameters

Property	Nomenclature	Value (units)	References
Material properties			
Melting temperature	T_m	2,324 (K)	[35–37]
Vaporization temperature	T_v	3,273.15 (K)	[35–37]
Latent heat of melting	L_m	1,067.43 (J/g)	[35–37]
Latent heat of evaporation	L_v	1,066.5 (J/g)	[35–37]
Mass of vapor molecule	M_v	1.693e–25 (kg/atm)	[35–37]
Thermal expansion coefficient	α	8.4e–6 (1/K)	[35, 36]
Temperature derivative of the surface tension	γ	–8e–5 (N/(m K))	[38]
Dynamic viscosity	μ	1 (Pa s for $T < 2,190$ K) and 0.001 (Pa s for $T > 2,500$ K)	[38]
Laser machining parameters			
Number of laser pulses	f	10, 20, 30, 40, 50 pulses	–
Pulse energy (E_p) and corresponding average laser energy density (E_d)	E_p $E_d = \frac{E_p}{(\frac{\pi}{4}D^2)} \times f$	1 (J) 3.5, 7.1, 10.6, 14.1, 17.7 (J/m ² × 10 ⁷), respectively	–
Pulse width	P_w	0.5 (ms)	–
Beam diameter	D	0.6 (mm)	–
Emissivity	ε	0.7	[1, 2]
Heat transfer coefficient	h	10 (W/m ² K)	[16]
Absorptivity	A	0.25	[38]
Half-width of the curve	ΔT	30 (K)	–
Reference point to represents the center of the laser beam	x_r	1,000 (μ m)	–
Standard deviation of the Gaussian laser beam	ϕ	100 (μ m)	–
Ambient temperature	T_a	293.15 (K)	–
Initial temperature	T_i	293.15 (K)	–
Stefan–Boltzmann constant	σ	5.67 × 10 ^{–8} (W/m ² K ⁴)	[36]
Universal gas constant	R	8.314 (J/mol K)	[36]
Gravitational acceleration	g	9.81 (m/s ²)	[36]

Table 2 Governing equations and boundary conditions of heat transfer model

Boundary no.	Boundary condition	Equation	No.
Whole geometry	Temperature-dependent specific heat at constant pressure (C_p)	$C_p = 880 + \delta_m(L_m) + \left(\frac{L_m}{T_m}\right) \cdot H'((T - T_m) \cdot \Delta T)$ $+ \delta_v(L_v) + \left(\frac{L_v}{T_v}\right) \cdot H'((T - T_v) \cdot \Delta T)$ (in J/kg K)	1
		$\delta_m = \frac{\exp\left[-\left(\frac{(T-T_m)^2}{\Delta T^2}\right)\right]}{\Delta T \sqrt{\pi}}, \quad \delta_v = \frac{\exp\left[-\left(\frac{(T-T_v)^2}{\Delta T^2}\right)\right]}{\Delta T \sqrt{\pi}}$	2 and 3
Whole geometry	Governing equation, step 1	$\rho C_p \left[\frac{\partial T}{\partial t}\right] = k \left[\left(\frac{\partial^2 T}{\partial x^2}\right) + \left(\frac{\partial^2 T}{\partial y^2}\right)\right]$	4
6	Heat flux, natural convection cooling and radiation	$-k \frac{\partial T}{\partial y} = \beta P_g - h[T - T_a] - \varepsilon \sigma [T^4 - T_a^4]$	5
	Average laser power density in Gaussian distribution	$P_g = A \left[\frac{E_p}{P_w \left(\frac{\pi}{4}D^2\right)}\right] \exp\left[-\left(\frac{(x-x_r)^2}{2\phi^2}\right)\right]$ (in W/m ²)	6
	Analytic function	$\beta = \text{rectl}\left(\text{mod}\left(t, \frac{1}{f}\right)\right)$	7
3, 8	Natural convection cooling and radiation	$-k \frac{\partial T}{\partial y} = h[T - T_a] - \varepsilon \sigma [T^4 - T_a^4]$	8
1, 9		$-k \frac{\partial T}{\partial x} = h[T - T_a] - \varepsilon \sigma [T^4 - T_a^4]$	9
2	Insulation	$\frac{\partial T}{\partial y} = 0$	10
4, 5, 7	Neutral	–	
Whole geometry	Level set method	$B = 1 \quad \text{for } (T > T_x + \Delta T)$	11
		$B = \frac{(T - T_x + \Delta T)}{2\Delta T} \quad \text{for } (T_x - \Delta T) \leq T \leq (T_x + \Delta T)$	12
		$B = 0 \quad \text{for } (T < T_x - \Delta T)$	13

Table 3 Governing equations and boundary conditions of fluid flow coupled with heat transfer model

Boundary no.	Boundary condition	Equation	No.
Whole geometry	Governing equations, step 2	$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$	14
		$\rho \left(\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} \right) = -\frac{\partial p}{\partial x} + \mu \left[\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right]$	15
		$\rho \left(\frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} \right) = -\frac{\partial p}{\partial y} + \mu \left[\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right] + \rho g \alpha (T - T_m)$	16
		$\rho C_p \left(\frac{\partial T}{\partial t} + u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} \right) = k \left[\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right]$	17
6	Recoil pressure	$P_r = (P_g) \frac{1.69}{\sqrt{L_v}} \left[\frac{\sqrt{\left(\frac{k T_s}{M_v L_v} \right)}}{1 + 2.2 \left(\frac{k T_s}{M_v L_v} \right)^2} \right]$ only when $T \geq T_v$	18
3, 6, 8	Surface tension	test $(u) \cdot \left[\frac{\partial \gamma}{\partial T} \frac{\partial T}{\partial x} \right]$	19
3, 6, 8	Open boundary	–	
Whole geometry	Temperature-dependent surface tension (γ) and dynamic viscosity (μ) for 2,190–2,500 K	$\gamma(T) = 0.64 - 8.2 \times 10^{-5} (T - T_m)$ $\mu(T) = 3.2 \cdot \left[\exp \left(\frac{43.2 \times 10^3}{RT} \right) \right]$	20 (21)

tension is at its weakest. Therefore, the fluid moves from the center to the edges of the molten pool.

4 Experimental approach

4.1 Laser machining

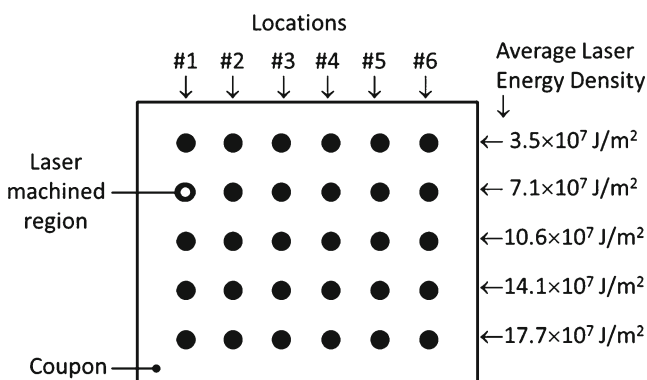
In the present study, the JK 701 Lumonics pulsed Nd:YAG (1.064 μm wavelength) fiber optic laser system is used to machine the alumina coupon ($50 \times 40 \times 6 \text{ mm}^3$; AdValue Technology Inc.). The alumina slab has a purity of 99.6 wt% (<0.1 % SiO_2 , <0.05 % Fe_2O_3 , and <0.1 % R_2O , all in weight percent). Table 1 summarized the laser machining parameters used in this study. In addition, Fig. 7 schematically represents the layout of laser-machined regions in alumina for various processing conditions.

4.2 Measurements of surface topography

Generally, there are various standard surface profile (topography) measurement techniques available, such as

contact stylus profilometer, contactless optical profilometer, scanning electron microscopy (SEM), and atomic force microscopy. However, the selection of appropriate measurement techniques is vital since it predominantly depends on the magnitude and periodicity of surface profiles. In the contact stylus tracing method, the main requirement for proper measurement is the size of the stylus tip and stem, which should be smaller than the roughness of the surface. On the contrary, the no-contact optical profilometer measurement is independent of the size of the stem and the stylus tip. Conversely, measuring the surface topography using SEM requires the cross-sectioning of the sample. However, it is extremely difficult to cross-section the ceramic sample (alumina) of finer dimensions (diameter and depth <1,000 μm) exactly across the center of the crater for accurate measurement. In addition, piled up and resolidified alumina around the crater lip tends to get removed during cross-sectioning and subsequent polishing, thereby leading to inaccurate surface profile (roughness) measurements. In light of this, the optical profilometer proved to be the most appropriate tool to conduct the topographic measurements on the surface and within the crater without missing the details of irregular pileup and resolidified layer around the crater lip. Hence, the optical profilometer is employed for surface profile measurements in the present work.

The NANOVEA[®] optical profilometer (model PS50) with an optical pen of 3.5 mm at the scan speed of 1 $\mu\text{m/s}$ is used for the surface profile measurements. The ISO 25178-2:2012 (Geometric Product Specifications—Surface Texture: Areal) specifications was followed during the surface profile measurements. In addition, the ISO 4287:1996 standards are utilized for the line profile measurement on the previously scanned surfaces to measure the amplitude parameters (R_t , R_p , and R_v). The raw data obtained during the surface profile measurements using the optical profilometer is further post-processed by using the NANOVEA[®]

**Fig. 7** Schematic of laser-machined alumina

3D software. In addition, the standard post-processing procedure suggested by the inbuilt NANOVEA® template is followed. The spatial filtering operator with a median denoising filter 3×3 is used to eliminate the outliers (bad points). The surface has been leveled by subtracting the least square plane from the whole surface area. Later, the median smoothing filter 7×7 was applied to filter out the noise (microroughness) and improve the quality of the scanned surface profile. Moreover, the surface was split into its waviness and roughness component by using an areal Gaussian filter with a cutoff of 0.25 mm. Finally, the line profile measurement was taken across the crater of the previously processed sample, as schematically shown in Fig. 8a. Thus, this procedure ensured the minimization or elimination of error in surface topography measurements. Further, a total of eight readings are taken for each processing condition ($8 \text{ line profiles} \times 3 \text{ locations} = 24 \text{ readings}$) in order to achieve higher accuracy in surface profile measurements. From each line profile measurement, the roughness parameters R_p , R_v , and R_t were evaluated (Fig. 8b).

5 Results and discussion

5.1 Prediction of material loss

From step 1, the temperature history for various laser machining conditions is extracted from boundary 6 (Fig. 9a). Due to higher laser energy density ($>3.5 \times 10^7 \text{ J/m}^2$) and the localized nature of the laser beam, the surface temperature in a confined volume quickly rises. Self-quenching and heat losses due to external natural convective cooling and radiation cause the drop in temperature. Both the increase and decrease of the temperature controls the different physical phenomena taking place in the material (heat conduction, melting, and vaporization). During one-dimensional multiple-pulse laser machining, the temperature rises when the laser is ON for a time duration of 0.5 ms (P_w) for each laser pulse, followed by the temperature drop during the laser OFF time ($[1/f] - P_w$). Based on the pulse rate ($f=10, 20, 30, 40, 50$), the laser is turned ON again for a second pulse (for duration of pulse width, $P_w=0.5 \text{ ms}$) which

increases the surface temperature. Likewise, the pulses are continuously delivered every $1/f$, in seconds, until the end of the laser cycle at 1 s causes the sequential rise and drop of temperature. In the present study, each individual laser pulse has constant laser pulse energy (1 J) and laser ON time ($P_w=0.5 \text{ ms}$). Due to the heat accumulation by the repetition of multiple laser pulses (Fig. 2b), the maximum surface temperature gradually increases with increase in number of laser pulses (Fig. 9a). But above a certain average laser energy density ($>7.1 \times 10^7 \text{ J/m}^2$), the material experiences the saturation state where the heat losses (natural convective cooling and radiation) are balanced by the given heat input (absorbed laser energy) and do not show substantial rise or drop of the surface temperature (Fig. 9a). However, the average laser energy density increases with increase in pulse rate which in turn increases the surface temperature (Fig. 9a). The average maximum temperature for all laser machining conditions are plotted to analyze the increase in surface temperature more accurately (Fig. 9b). It is also observed that the average surface temperature increases with increase in pulse rate (or average laser energy density). Even for lowest average laser energy density ($3.5 \times 10^7 \text{ J/m}^2$) used in the current study (Fig. 9b), the average maximum temperature (4,065 K) is considerably higher than vaporization temperature (3,273 K), indicating the higher material loss due to evaporation.

Due to the rapid increase and decrease of surface temperature (Fig. 9a), the material experiences the higher heating and cooling rates (dT/dt) during laser machining. Furthermore, due to repetition of multiple laser pulses on the alumina surface, the material initially experiences a sudden increase in temperature which is evidenced by the initial higher heating/cooling rates (Fig. 10a). However, once the bulk material attains the saturation state, the further increases in temperature in subsequent laser pulses are not as high as in the initial stage and caused the gradual drop in heating/cooling rates until the laser is switched off. As a result, the heating/cooling rate decreases with increase in number of pulses for all laser machining conditions (Fig. 10a). The average value of heating/cooling rate is calculated to represent the corresponding value of heating/cooling rate for each machining condition (Fig. 10b). It can be observed that the average heating/cooling rate decreases with increase in average laser energy density (or pulse rate) for all laser machining conditions (Fig. 10b). Furthermore, it is noticed that, as the pulse rate increases, the material remains in the liquid phase for longer time and the surface deforms continuously during this time.

The level set method is used to predict the crater depth and width for various laser machining conditions. The crater depth and width increases with the increase in average laser energy density or pulse rate (Fig. 11). In the present work, the energy distribution of laser beam is in Gaussian, i.e., the

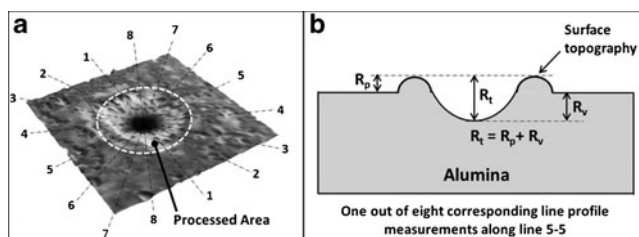
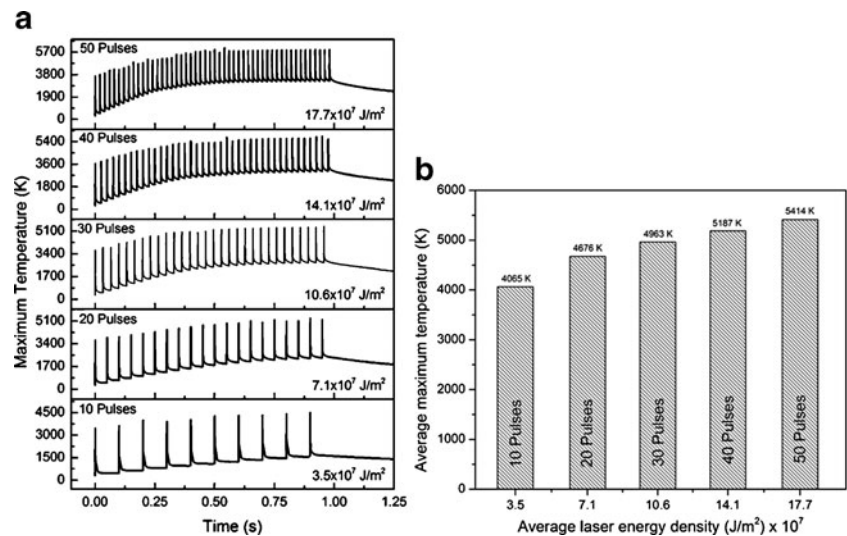


Fig. 8 Surface profile measurement: **a** tracks of line profile measurements and **b** two-dimensional representation of line profile measurements along line 5–5 and corresponding roughness parameters

Fig. 9 Computational plots: **a** maximum temperature as a function of time and **b** average maximum temperature as a function of average laser energy density



laser energy is highest at the center of the beam and it gradually reduces as it goes towards the edges of the laser beam. Therefore, the surface temperature is higher at the center of the laser beam compared to the edge. As a result, the material experiences preferential heat conduction. Hence, the penetration of heat inside the material normal to the top surface is higher than the lateral direction, causing the higher crater depth compared to crater width (Fig. 11). During the repetition of multiple laser pulse, the material experiences a sequential loss of material due to evaporation. This material lost also increases with increase in number of laser pulses, which directly contributes in increasing the crater depth. From this data, the updated geometry was then created and utilized in step 2.

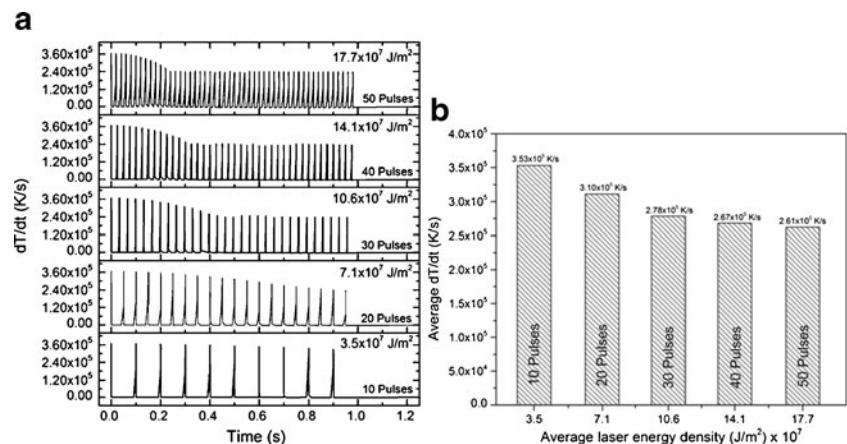
5.2 Prediction of surface topography

The fluid flow coupled with heat transfer and phase change kinetics is incorporated in step 2. Based on the dimension of crater depth and width provided by step 1 heat transfer model, the new geometry (creation of crater) is designed in step 2 to include the material loss due to evaporation. It

can be noticed that the shape of boundary 6 (Fig. 4) in step 1 is flat, but due to the loss of material, it had changed to concave shape. Thus, the new surface geometry of boundary 6 precisely defines the normal and tangential boundary conditions (heat flux, recoil pressure, and surface tension). Even though the heat transfer and phase change boundary conditions adopted in step 2 are the same as step 1, the change in surface geometry of boundary 6 (Fig. 5) strongly influences the corresponding boundary conditions of fluid flow, heat transfer, and phase change kinetics and, hence, the generation of final surface profile.

The recoil pressure, surface tension, and gravitational forces are the important factors to generate the hydrodynamic melt pool motion. During the heating cycle, when the surface temperature is above the vaporization temperature, the recoil pressure is the dominating factor to drive the molten material out from the crater that further increases the crater depth as well as the liquid pileup. On the other hand, when the temperature drops below the vaporization temperature, surface tension and gravitational forces lead to the movement of the molten material and, therefore, liquid metal tends to return to its original place. At the end of the

Fig. 10 Computational plots: **a** rate of heating/cooling (dT/dt) as a function of time and **b** average dT/dt as a function of average laser energy density



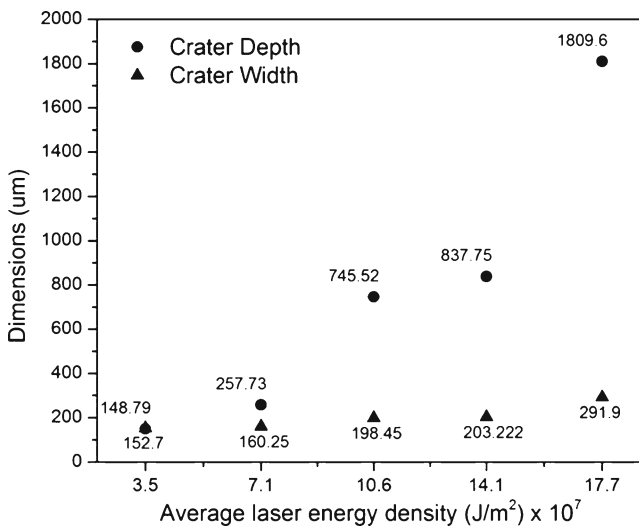


Fig. 11 Measured crater depth and width for various laser machining conditions

heating cycle, the higher cooling rates and self-quenching that simultaneously affect the solidifying liquid material and the tangential stresses generated by the surface tension forces provide the profile of the solidified material.

The recoil pressure strongly depends on average laser energy density and the instantaneous surface temperature (Eq. 18) and, hence, its magnitude gradually increases with increase in average laser energy density and the instantaneous surface temperature (Fig. 12). Figure 12 only shows the numerically calculated recoil pressure during various laser energy density machining conditions.

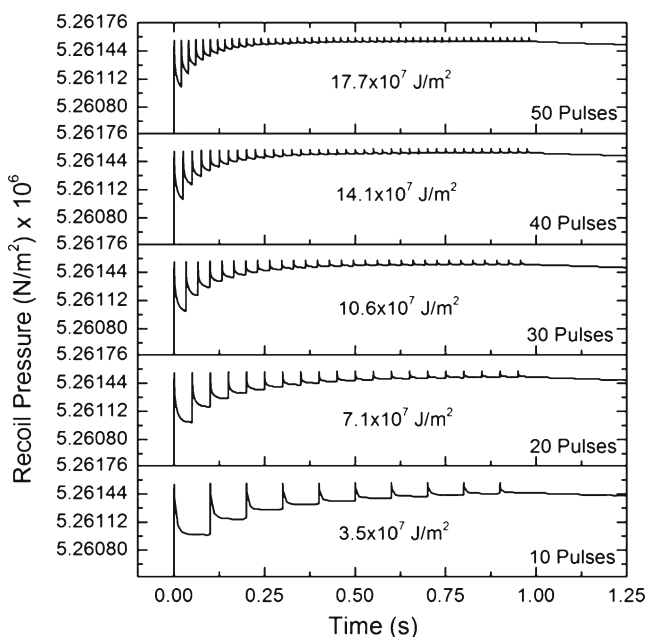


Fig. 12 Effect of recoil pressure as a function of time for various laser energy density machining conditions

However, in the computational model, the actual recoil pressure is only applied on the melt pool when the surface temperature reaches above the vaporization temperature. Once the material reaches the saturation state, the further increase in the number of pulses or average laser energy density does not alter the magnitude of recoil pressure (Fig. 12). In addition, the depth of the crater increases drastically due to the increase in average laser energy density (or pulse rate) and recoil pressure. However, above a certain critical crater depth ($\sim 260 \mu\text{m}$), the magnitude of recoil pressure is not enough to drive the molten material out of the crater even though the surface temperature is above the vaporization temperature. Here, the magnitude of surface tension and gravitational forces dominate over the higher recoil pressure, causing the drop in velocity of molten material. Hence, net velocity of the molten material initially increases with increase in average laser energy density (3.5×10^7 and $5.3 \times 10^7 \text{ J/m}^2$) (Fig. 13). However, after reaching a certain critical crater depth ($\sim 260 \mu\text{m}$), net velocity of molten material drops down for further increase in average laser energy density ($> 5.3 \times 10^7 \text{ J/m}^2$). The velocity is high at the end of the laser on time due to the maximum temperature and corresponding maximum recoil pressure for all the average laser energy density machining conditions that existed at the end of the heating cycle (Fig. 13). However, as the surface temperature continuously drops down immediately after the laser off time until the end of the cycle (1 s), it in turn causes the gradual drop in the velocity of the molten material.

The computationally predicted surface deformation plots in cross-section for various laser machining conditions are presented in Fig. 14a. The present computational model assumes that the piled up and/or resolidified material is symmetric around the plane $x-x$ (Figs. 2c and 14b). In this

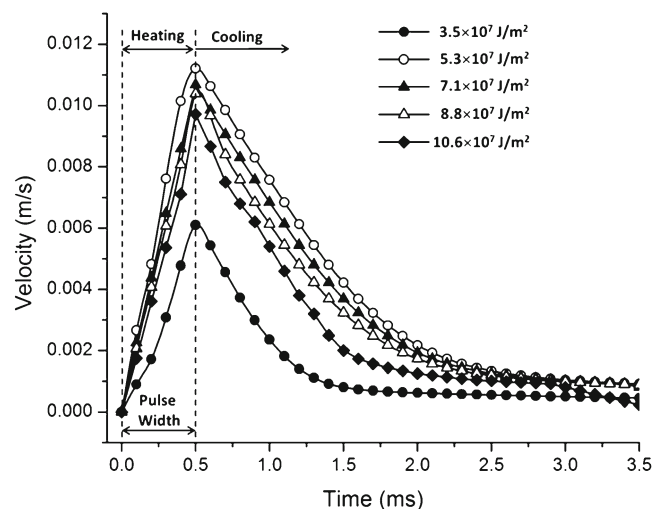


Fig. 13 The velocity profile for various laser energy density machining conditions during the last laser pulse

case, the predicted results are revolved around the axis (z-axis, Fig. 14c) of the crater (along the circumference) and, hence, a uniform ring-like structure formed along the crater lip, as illustrated in Fig. 14d. Under this assumption, the cross-sectional view (Fig. 14a) facilitates the measurement of the surface roughness parameters R_p , R_v , and R_t . During the application of each laser pulse, the surface temperature of the material goes above the vaporization temperature, leading to material loss due to evaporation. As the number of laser pulses increases, the material experiences material removal which increases the crater depth. In addition to this, the molten material expelled out by recoil pressure directly

contributes to increasing the R_v value (Figs. 14 and 15). Although the crater depth increases, the recoil pressure is not sufficient to eject the molten material out from the crater, resulting in a decrease in the R_p value (Figs. 14 and 15). Hence, after reaching a certain critical crater depth ($\sim 260 \mu\text{m}$), the net velocity of the molten material decreases with increase in average laser energy density. On the contrary, a longer time necessary for solidification ($\sim 2\text{--}5 \text{ s}$) allows the expelled molten material to solidify on the inside wall of the crater (known as recast layer) without the possibility of returning to its original place which will reduce the pileup height (R_p) (Figs. 8b, 14, and 15). Thus, distinct

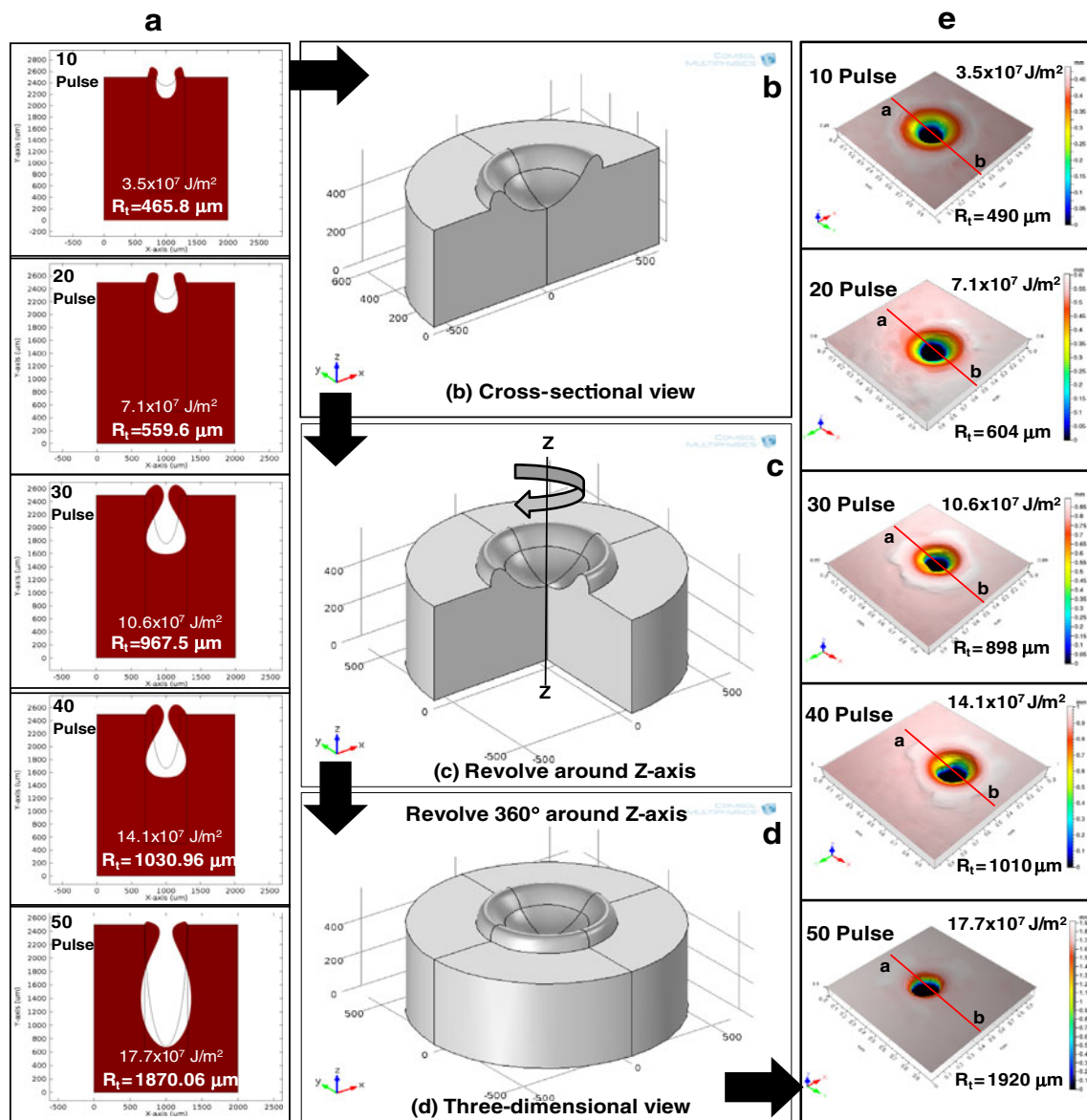


Fig. 14 Computationally predicted and experimentally measured surface topography (profiles): **a** computationally predicted surface topography, **b** schematic cross-sectional view of predicted surface topography, **c** schematic of predicted surface profile revolved around

the z-axis, **d** schematic three-dimensional view of predicted surface topography, and **e** experimentally measured surface topography (ISO 25178) and one out of eight corresponding line profile measurements along line a–b (ISO 4287)

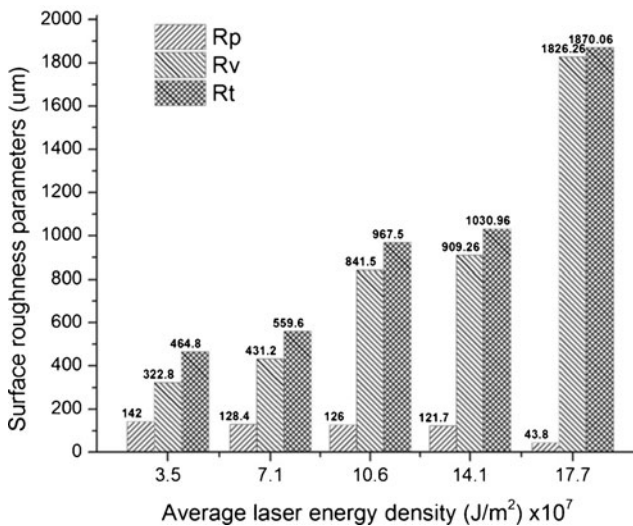


Fig. 15 The change of surface roughness parameters for various average laser energy density machining conditions

teardrop shape topography is generated on the wall of the crater (Fig. 14a). Among all laser machining conditions, the highest laser energy density machining condition ($10.6 \times 10^7 \text{ J/m}^2$) provides the largest crater depth ($R_v = 1826.26 \text{ μm}$) with the least material pileup ($R_p = 43.8 \text{ μm}$) on the surface. This combination is responsible for the higher surface roughness ($R_t = 1870.06 \text{ μm}$) for higher energy density.

5.3 Validation of computational model

In order to validate the computational model, the computationally predicted and experimentally measured values of surface roughness for various laser machining conditions are summarized and presented in Table 4. It is observed that the surface roughness increases with the increase in average laser energy density. The difference between the computational model and the experimental observations are within the range of 2–8 %, thereby strongly validating the computational model. The difference of computational results can be attributed to the adaptation of various literature-suggested temperature-

dependent material properties [36–38] and boundary conditions [1, 2, 34, 42, 43].

The present computational model incorporates temperature-dependent material properties, such as thermal conductivity, density, heat capacity, absorptivity, viscosity, and surface tension coefficient. The deviation of these thermophysical properties may play a critical role in generating the hydrodynamic melt motion of the liquid metal during the simulation. Furthermore, during laser processing, the crater shape is also constantly changing, which can vary the angle of incidence of the laser beam. Therefore, the change in absorbance of the laser energy can cause the variation in surface temperature and, consequently, the temperature-dependent material properties. Hence, the more accurate thermophysical properties can further reduce the difference between computational and experimental values. Moreover, as mentioned earlier, the in situ measurement of variable thermophysical properties is a very challenging task due to high temperature, very short duration, and very small volume of laser-material interaction. Therefore, very limited data are presently available in the open literature. In addition, efforts are underway to minimize the difference between the results of computational model and experimental observations.

6 Conclusion

COMSOL™ Multiphysics was used to develop a computational model to understand the influence of multiple laser pulses on the surface finish of alumina for various laser energy density machining conditions. Results indicate that the material lost due to evaporation causes an increase in crater depth, whereas liquid expulsion created by the recoil pressure increases the pileup height. However, after a critical crater depth ($\sim 260 \text{ μm}$), the magnitude of the recoil pressure was insufficient to eject the significant amount of liquid material out of the crater and hence the liquid material solidified inside the crater wall, leading to the formation of a typical teardrop

Table 4 Computationally predicted and experimentally measured surface roughness

Experiment	Parameters				
	Average energy density (J/m^2)	Laser pulses (1/s)	Numerically calculated R_t (μm)	Experimentally measured $R_t \pm \text{standard deviation}$ (μm)	Difference (%)
1	3.5×10^7	10	465.8	490 ± 5.92	5.19
2	7.1×10^7	20	559.6	604 ± 3.80	7.93
3	10.6×10^7	30	967.5	898 ± 5.16	7.18
4	14.1×10^7	40	1,030.96	$1,010 \pm 4.84$	2.03
5	17.7×10^7	50	1,870.06	$1,920 \pm 3.89$	2.67

shape topography. In this study, it was found that the surface roughness increased with increasing pulse rate (10, 20, 30, 40, and 50) or the increase in average laser energy density. A close agreement and a similar trend were observed between the predicted roughness values of the computational model with experimental observations.

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References

- Samant AN, Dahotre NB (2009) Laser machining of structural ceramics—a review. *J Eur Ceram Soc*. doi:10.1016/j.jeurceramsoc.2008.11.010
- Samant AN (2009) Laser machining of structural ceramics: computational and experimental analysis. Dissertation or Thesis, University of Tennessee
- Tuersley IP, Jawaid A, Pashby IR (1994) Review: various methods of machining advanced ceramic materials. *J Mater Process Technol*. doi:10.1016/0924-0136(94)90144-9
- Vikulin VV, Kelina IY, Shatalin AS, Rusanova LN (2004) Advanced ceramic structural materials. *Refract Ind Ceram*. doi:10.1007/s11148-005-0017-2
- Shen JY, Luo CB, Zeng WM, Xu XP, Gao YS (2002) Ceramics grinding under the condition of constant pressure. *J Mater Process Technol*. doi:10.1016/S0924-0136(02)00636-2
- König W, Wagemann A (1993) Machining of ceramic components—process-technological potentials. National Institute of Standards and Technology (NIST): Machining of Advanced Materials Proceedings. 45:3–16
- Chryssolouris G, Anifantis N, Karagiannis S (1997) Laser assisted machining: an overview. *J Manuf Sci Eng Trans ASME* 119:766–769
- Kacar E, Mutlu M, Akman E, Demir A, Candan L, Canel T, Gunay V, Simmazcelik T (2009) Characterization of the drilling alumina ceramic using Nd:YAG pulsed laser. *J Mater Process Technol*. doi:10.1016/j.jmatprotec.2008.04.049
- Harimkar SP, Dahotre NB (2006) Effect of laser fluence on surface microstructure of alumina ceramic. *Advances in Applied Ceramics* 105:304–308
- Triantafyllidis D, Li L, Stott FH (2002) Surface treatment of alumina-based ceramics using combined laser sources. *Appl Surf Sci*. doi:10.1016/S0169-4332(01)00639-0
- Koepke BG, Stokes RJ (1970) A study of grinding damage in magnesium oxide single crystals. *J Mater Sci*. doi:10.1007/BF00551000
- Kirchner HP (1984) Damage penetration at elongated machining grooves in hot-pressed Si_3N_4 . *J Am Ceram Soc*. doi:10.1111/j.1151-2916.1984.tb09629.x
- Conway JC, Kirchner HP (1986) Crack branching as a mechanism of crushing during grinding. *J Am Ceram Soc*. doi:10.1111/j.1151-2916.1986.tb04815.x
- Dahotre NB, Samant AN (2011) Laser machining of advanced materials. CRC, Boca Raton
- Dahotre NB, Harimkar SP (2008) Laser fabrication and machining of materials. Springer, New York
- Kannatey-Asibu E (2009) Principles of laser materials processing. Wiley, Hoboken
- Chryssolouris G (1991) Laser machining: theory and practice. Springer, New York
- Salonitis K, Stavropoulos P, Stourmaras A, Chryssolouris G (2007) CO_2 laser cutting of aluminium. Proceedings of the 5th Laser Assisted Net Shape Engineering pp 825–835
- Stavropoulos P, Salonitis K, Stourmaras A, Euthimiou K, Chryssolouris G (2007) Molecular dynamics simulation of laser ablation of iron. Proceeding of the 10th CIRP international workshop on modeling of machining operations, pp 549–553
- Stourmaras A, Salonitis K, Stavropoulos P, Chryssolouris G (2007) Finite element thermal analysis of pulsed laser drilling process. Proceeding of the 10th CIRP international workshop on modeling of machining operations, pp 563–570
- Dubey AK, Yadava V (2008) Laser beam machining—a review. *Int J Mach Tools Manuf*. doi:10.1016/j.ijmachtools.2007.10.017
- Yeo CY, Tam SC, Jana S, Lau MWS (1994) A technical review of the laser drilling of aerospace materials. *J Mater Process Technol*. doi:10.1016/0924-0136(94)90073-6
- Corcoran A, Sexton L, Seaman B, Ryan P, Byrne G (2002) The laser drilling of multi-layer aerospace material systems. *J Mater Process Technol*. doi:10.1016/S0924-0136(01)01123-2
- Li L, Diver C, Atkinson J, Giedl-Wagner R, Helml HJ (2006) Sequential laser and EDM micro-drilling for next generation fuel injection nozzle manufacture. *CIRP Ann Manuf Technol*. doi:10.1016/S0007-8506(07)60393-X
- Lawrence J, Low DKY, Pou J, Toyserkani E (2010) Advances in laser materials processing technology. Woodhead Publishing Ltd., Cambridge
- Hanon MM, Akman E, Genc Oztoprak B, Gunes M, Taha ZA, Hajim KI, Kacar E, Gundogdu O, Demir A (2012) Experimental and theoretical investigation of the drilling of alumina ceramic using Nd:YAG pulsed laser. *Opt Laser Technol*. doi:10.1016/j.optlastec.2011.11.010
- Samant AN, Dahotre NB (2008) Computational predictions in single-dimensional laser machining of alumina. *Int J Mach Tools Manuf*. doi:10.1016/j.ijmachtools.2008.05.004
- Samant AN, Dahotre NB (2009) Differences in physical phenomena governing laser machining of structural ceramics. *Ceram Int*. doi:10.1016/j.ceramint.2008.11.013
- Harimkar SP, Samant AN, Dahotre NB (2007) Temporally evolved recoil pressure driven melt infiltration during laser surface modifications of porous alumina ceramic. *Journal of Applied Physics*. doi:10.1063/1.2710288. http://jap.aip.org/resource/1/japiau/v101/i5/p054911_s1
- Sun S, Durandet Y, Brandt M (2005) Parametric investigation of pulsed Nd:YAG laser cladding of stellite 6 on stainless steel. *Surf Coat Technol*. doi:10.1016/j.surfcoat.2004.03.058
- Islama MU, Campbella G (1993) Laser machining of ceramics: a review. *Materials and Manufacturing Processes*. doi:10.1080/10426919308934870
- Vora HD, Santhanakrishnan S, Harimkar SP, Boetcher SKS, Dahotre NB (2012) Evolution of surface topography in one-dimensional laser machining of structural alumina. *J Eur Ceram Soc*. doi:10.1016/j.jeurceramsoc.2012.06.015
- Anisimov SI (1968) Vaporization of metal absorbing laser radiation. *Soviet Physics J Exp Theor Phys* 27:182
- Semak V, Matsunawa A (1997) The role of recoil pressure in energy balance during laser materials processing. *J Phys D Appl Phys* 30:2541–2552
- Quintero F, Varas F, Pou J, Lusquinos F, Boutinguiza M, Soto R, Pérez-Amor M (2005) Theoretical analysis of material removal mechanisms in pulsed laser fusion cutting of ceramics. *J Phys D Appl Phys* 38:655–666
- Kolev NI (2011) Multiphase flow dynamics 5: nuclear thermal hydraulics. Springer, Berlin

37. Gitzen WH (1970) Alumina as a ceramic material. American Ceramic Society, Columbus
38. Paradis P, Ishikawa T (2005) Surface tension and viscosity measurement of liquid and undercooled alumina by contactless techniques. *Jpn J Appl Phys* 44:5082–5085
39. Samant AN, Du B, Dahotre NB (2009) In-situ surface absorptivity prediction during 1.06 μm wavelength laser low aspect ratio machining of structural ceramics. *Physica Status Solidi A*. doi:[10.1002/pssa.200925108](https://doi.org/10.1002/pssa.200925108)
40. COMSOL (2011) Continuous casting model. COMSOL Multiphysics model library V 4.2a
41. COMSOL (2011) Users guide. COMSOL Multiphysics model library V 4.2a
42. Sussman M, Smereka P, Osher S (1994) A level set approach for computing solutions to incompressible two-phase flow. *J Comput Phys*. doi:[10.1006/jcph.1994.1155](https://doi.org/10.1006/jcph.1994.1155)
43. COMSOL (2011) Marangoni convection model. COMSOL Multiphysics model library V 4.2a
44. Pierron N, Sallamand P, Mattei S (2005) Numerical modeling of molten pool formation during an interaction of a pulse laser (Nd: YAG) with an aluminum sheet. *Proceedings of the COMSOL Multiphysics User's Conference*, pp 1–5